PYTHON ACTIVITY-BASED LEARNING

Case Study 03: Repeated Transaction Detection

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Student Name: _____ **Roll No:** _____

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Problem Statement

Process transaction IDs and amounts, detect repeated IDs and flag transactions above a specified amount.

Objectives

• Store transaction IDs and amounts in lists.

• Find duplicate transaction identifiers.

• Identify transactions above a threshold.

Assumptions

• Transaction IDs are strings.

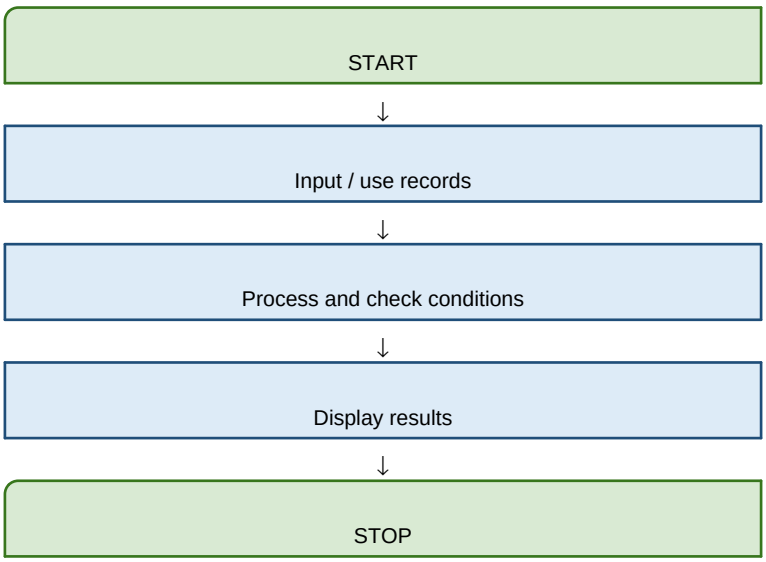
• An amount greater than 5000 is considered high value.

• The sample transaction list represents one day's transactions.

Algorithm

1. Store transaction records in a list of tuples.
2. Create an empty list to track seen IDs.
3. For each transaction, check whether its ID is already seen.
4. Report duplicate IDs and flag amounts greater than 5000.
5. Display the suspicious transaction count.

Flowchart



Pseudocode

```
START
SET seen IDs to empty list
FOR each transaction
    IF ID is already in seen IDs, DISPLAY duplicate
    ELSE ADD ID to seen IDs
    IF amount > 5000, DISPLAY high-value transaction
STOP
```

Python Source Code

```
transactions = [
    ("T101", 1200),
    ("T102", 6500),
    ("T103", 900),
    ("T102", 3000),
    ("T104", 7200)
]

seen = []
print("Suspicious transactions:")

for tid, amount in transactions:
    if tid in seen:
        print(tid, "Duplicate ID")
    else:
        seen.append(tid)

    if amount > 5000:
        print(tid, "High amount:", amount)
```

Test Cases

Test input / condition	Expected result
T102 appears twice	T102 is reported as duplicate
Amount = 6500	Reported as high amount
Amount = 900	Not flagged by threshold

Output Screenshot

Sample console output (formatted to show the expected execution result):

```
Suspicious transactions:
T102 High amount: 6500
T102 Duplicate ID
T104 High amount: 7200
```

Conclusion

The program detects repeated identifiers and high-value transactions using lists and conditional statements.